The controversy finally died in the 1930s, as technology became sufficiently advanced to produce accurate experimental verifications of special relativity's predictions. By now, in the 1990s, there is absolutely no room for doubt: Every day more than 10^{17} electrons in particle accelerators at Stanford University, Cornell University, and elsewhere are driven up to speeds as great as 0.9999999995 of the speed of light—and their behaviors at these ultra-high speeds are in complete accord with Einstein's special relativistic laws of physics. For example, the electrons' inertia increases as they near the speed of light, preventing them from ever reaching it; and when the electrons collide with targets, they produce high-speed particles called mu mesons that live for only 2.22 microseconds as measured by their own time, but because of time dilation live for 100 microseconds or more as measured by the physicists' time, at rest in the laboratory.

The Nature of Physical Law

Does the success of Einstein's special relativity mean that we must totally abandon the Newtonian laws of physics? Obviously not. The Newtonian laws are still used widely in everyday life, in most fields of science, and in most technology. We don't pay attention to time dilation when planning an airplane trip, and engineers don't worry about length contraction when designing an airplane. The dilation and contraction are far too small to be of concern.

Of course, if we wished to, we *could* use Einstein's laws rather than Newton's in everyday life. The two give almost precisely the same predictions for all physical effects, since everyday life entails relative speeds that are very small compared to the speed of light.

Einstein's and Newton's predictions begin to diverge strongly only at relative speeds approaching the speed of light. Then and only then must one abandon Newton's predictions and adhere strictly to Einstein's.

This is an example of a very general pattern, one that we shall meet again in future chapters. It is a pattern that has been repeated over and over in the history of twentieth-century physics: One set of laws (in our case the *Newtonian laws)* is widely accepted at first, because it accords beautifully with experiment. But then experiments become more accurate and this first set of laws turns out to work well only in a limited domain, its *domain of*